

Kronecker Factor

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We want to prove the following result generalised for amenable groups, where T is replaced with Γ :

Proposition 0.1 (Host, 2019, Proposition 5). *Let (X, T) be a topological dynamical system, $x_0 \in X$, and μ be an ergodic invariant probability measure supported on the closed orbit of x_0 under T . Let $((Z, m_Z, S), m_Z, S)$ be the Kronecker factor of (X, μ, T) , with factor map $p : X \rightarrow Z$. Let $X \times Z$ be endowed with the transformation of $T \times S$. Let $\tilde{\mu}$ be the measure on $X \times Z$ and image of μ under the map $X \rightarrow X \times Z$ where $x \mapsto (x, p(x))$.*

Then there exists a Følner sequence $\tilde{\Phi}$ and a point $z_0 \in Z$ such that (x_0, z_0) is generic for $\tilde{\mu}$ along $\tilde{\Phi}$.

Host (2019) also concludes that, for every $f \in \mathcal{C}(X)$ and every $h \in \mathcal{C}(Z)$, we have

$$\int_X f(x) \cdot (h \circ p)(x) = \lim_{N \rightarrow \infty} \frac{1}{|\tilde{\Phi}_N|} \sum_{n \in \tilde{\Phi}_N} f(T^n x_0) h(S^n z_0).$$

1 Background

Firstly, we have the definitions of the Kronecker subsystem and Kronecker factor below:

Definition 1.1 (Jamneshan and Kreidler, 2025, Definition 7.1.14 & Subsection 13.3). Let (X, T) be a measure-preserving system. We then write $J_{\text{kro}} : (X_{\text{kro}}, T_{\text{kro}}) \rightarrow (X, T)$ for the extension $J_E : (X_E, T_E) \rightarrow (X, T)$ defined by the invariant Markov sublattice $E = L^2(X)_{\text{ds}}$ and call $(X_{\text{kro}}, T_{\text{kro}})$ the **Kronecker subsystem** of (X, T) .

We refer to $(X_{\text{kro}}, \tau_{\text{kro}})$ as the *Kronecker factor* of the system (X, τ) .

Kra and Host (2007, Proposition 6.1) and Host and Kra (2018, Proposition 24.3) were the sources given for Host (2019, Proposition 5). We will compare these original results with the result we intend to prove.

Proposition 1.1 (Kra and Host, 2007, Proposition 6.1). *Let (X, T) be a topological dynamical system, $x_0 \in X$ a transitive point, and μ an invariant*

ergodic measure on X . Let (Y, S) be a distal topological dynamical system, ν an invariant measure on Y , and $p : (X, \mu, T) \mapsto (Y, \nu, S)$ a measure theoretic factor map.

Then there exists a point $y_0 \in Y$ and a sequence of intervals $I = (I_j : j \geq 1)$ whose lengths tend to infinity such that for every continuous function f on X and every continuous function g on Y ,

$$\int f(x)(g \circ p)(x) \, d\mu(x) = \lim_{j \rightarrow +\infty} \frac{1}{|I_j|} \sum_{n \in I_j} f(T^n x_0) g(S^n y_0).$$

Comparing the assumptions for Proposition 0.1 with Proposition 1.1, we have:

	Proposition 0.1	Proposition 1.1
1		$x_0 \in X$ has been specified as a transitive point.
2	μ has been specified as being supported on the closed orbit of x_0 under the action of T .	
3	(Y, S) is defined as the Kronecker factor (Z, m_Z, S) .	

Additionally:

- $\tilde{\mu}$ appears to be a joining between μ and ν .
- I is the Følner sequence $\tilde{\Phi}$.

Left to check definitions for:

- (Kra and Host, 2007, Theorem 3.1): Let $k \geq 1$ be an integer. For a k -step nilsystem, $(X = G/\Gamma, T)$ with Haar measure μ .
 - G is a k -step nilpotent Lie group
 - * G is a group where, for $g, h \in G$, we write $[g, h] = ghg^{-1}h^{-1}$ for the commutator of g and h and we write $[A, B]$ for the subgroup spanned by $\{[a, b] : a \in A, b \in B\}$.
 - * The commutator subgroups G_j , $j \geq 1$, are defined inductively by setting $G_1 = G$ and $G_{j+1} = [G_j, G]$.
 - * Let $k \geq 1$ be an integer. We say that G is a k -step nilpotent if G_{k+1} is the trivial subgroup.
 - Γ is a discrete cocompact subgroup of G
 - The compact manifold $X = G/\Gamma$ is called a k -step nilmanifold
 - * The group G acts on X by left translations and we write this action as $(g, x) \mapsto g.x$.
 - (X, T) is *transitive* (admits a dense orbit).
 - * This is equivalent to being minimal, uniquely ergodic, or (X, μ, T) being ergodic.

- [distal]
 - previously seen def
 - * A topological system (X, T) is **distal** if

$$\inf\{d(T^n x, T^n y) : n \in \mathbb{N}\} > 0$$

- for all $x \neq y \in X$, where d is the metric on X . (Kra et al., 2022)
- a def exists for any semigroup action by assuming there's a metric compact space
- a def exists for equicontinuous
- Discuss why has distal systems been used?
 - * Is it required for the application of the result?
 - * Is it required for the proof?
- check the Kronecker system in my case is distal

Kra and Host, 2007, p. 257. [Section 4.1 (p. 24) - association of (X, T) and x_0 to a sequence (the start is just FCP)]

Section 4.1:

- All FCP?
 - Define: separable subalgebra of $\ell^\infty(\mathbb{Z})$
 - * Separable subalgebra of $\ell^\infty(\mathbb{Z})$ is meant as unitary subalgebra of $\ell^\infty(\mathbb{Z})$ as well as (*)
 - ISem28: An amenable group, Γ , has a group homomorphism $U : \Gamma \rightarrow \mathcal{U}(H)$ where $\gamma \mapsto U_\gamma$ is a unitary representation of Γ on H . The set of all unitary operators, $\mathcal{U}(H)$, is the subset of the set of linear isometries $U : H \rightarrow H$ (such that $\|Uf\| = \|f\|$ for all $f \in H$) that are surjective (and thus bijective).
 - Would the amenable case of the separable subalgebra be the case where $f = U_\gamma \circ r$ where $r \in \ell^\infty(X)$?
 - Kra and Host (2007) is using a functional analysis approach to FCP [Attempt following the steps with the FCP we covered].
 - For $\ell^\infty(\Gamma)$, a could be $(1_\gamma : \gamma \in \Gamma)$.
 - Easier to think of $\ell^\infty(\mathbb{Z})$ as all the bounded functions on $\mathbb{Z}$ and has the supremum norm. For $B \subset \mathbb{Z}$ you can define a function $f \in \ell^\infty(\mathbb{Z})$, i.e., 1_B .
 - * (*):
 - invariant under the shift
 - invariant under complex conjugation
 - closed in $\ell^\infty(\mathbb{Z})$
 - [separable] for the uniform norm $\|\cdot\|_\infty$
 - * We mostly consider the case of the separable subalgebra $\mathcal{A}(a)$ spanned by a bounded sequence $a = (a_n : n \in \mathbb{Z})$.
 - * The shift on $\ell^\infty(\mathbb{Z})$ is written as σ and thus, for a sequence $a = (a_n : n \in \mathbb{Z})$, $\sigma a = (a_{n+1} : n \in \mathbb{Z})$

* We use $\bar{a}$ to denote the conjugate sequence $(\bar{a} : n \in \mathbb{Z})$.

[If Y denotes the Kronecker factor of (X, μ, T)]

[The sequence of intervals I plays the same role as the “Kronecker complete processes” of [BFW]]

We write $d_X(\cdot, \cdot)$ and $d_Y(\cdot, \cdot)$ for distances on X and Y , defining the topologies of these spaces.

Subsection 6.1.1 (p. 39)

- $\mathcal{B}$ closed subalgebra of $L^\infty(\mu)$ spanned by $\mathcal{C}(X)$ and the function $g \circ \pi$ with $g \in \mathcal{C}(Y)$.
 - It is unitary, separable, and invariant under complex conjugation and under T .
- W is the Gelfand spectrum of this algebra.
 - Let W be the Gelfand spectrum of $\mathcal{B}$ then W consists of the set of unitary homomorphisms from $\mathcal{B}$ to the complex numbers.
 - * W will be a “nice” space - compact metric
 - * Anything we have on $\mathcal{B}$ will have analogies in W
 - As $\mathcal{B}$ is invariant under T , then an equivalent invariance is in W , see $R : W \rightarrow W$.
 - Like how $Tf = f \circ T$ looks at pointwise transformations on $\mathcal{B}$, this goes a step further with R on W .
 - * Can construct the Kronecker factor with the Gelfand spectrum.
 - This would require a $\mathcal{B}$ generated by the eigenfunctions, which relates to our previous work on assuming an Sigma Algebra of almost-invariant subsets.
 - Compare with ISem28 Exercise 8.5.
 - This method differs to using disintegrations as disintegrations start with assuming a sigma algebra of almost-invariant subsets whereas Gelfand spectrums starts with all the eigenfunctions (which may not completely correspond to almost-invariant subsets) and create a sigma algebra from this (thus being more general).
 - As $\mathcal{B}$ is separable, W is a compact metrizable space.
 - By def, there exists an isometric isomorphism of algebras $\Psi : \mathcal{C}(W) \rightarrow \mathcal{B}$.
 - * For $b \in \mathcal{B}$, the function $\Psi^{-1}(b)$ is called *the function associated to b* .
 - * The map $b \mapsto b_0$ is a character of the algebra $\mathcal{B}$. Thus, there exists a point $w_0 \in W$ with $\Psi^{-1}(b) = f \mapsto f(w_0)$ for all $f \in \mathcal{C}(W)$
- The shift on $\mathcal{B}$ induces a homeomorphism $R : W \rightarrow W$ satisfying $\Psi(f \circ T) = \Psi(f) \circ R$ for all functions $f \in \mathcal{C}(W)$.
 - Therefore, for every $f \in \mathcal{C}(W)$, $\Psi(f)$ is the sequence

$$\Psi(f) = (f(T^n w_0) : n \in \mathbb{Z}).$$

- If $f \in \mathcal{C}(W)$ satisfies $f(T^n w_0) = 0$ for all $n \in \mathbb{Z}$, then the sequence given by $\Psi(b) = f$ is identically zero and so f itself is identically zero.
- * It follows that the point w_0 of W is transitive, meaning that its orbit $\{T^n w_0 : n \in \mathbb{Z}\}$ is dense in W .
- The triple (W, T, w_0) is called the *pointed topological dynamical system associated to the algebra $\mathcal{B}$* .
- The inclusion of $\mathcal{C}(X)$ in $\mathcal{B}$ induces a continuous surjective map $p : W \rightarrow X$ satisfying $f \circ p = \Psi(f)$ for every continuous function f on X and we have that $T \circ p = p \circ R$.
 - Similarly, the map $g \mapsto g \circ \pi$ from $\mathcal{C}(Y)$ to $\mathcal{B}$ is an isometric homomorphism of algebras and thus induces a continuous surjective map $q : W \rightarrow Y$ satisfying $g \circ q = \Psi(g \circ \pi)$ for all continuous functions g on Y .
 - We have that $S \circ q = q \circ R$ so $p : (W, R) \rightarrow (X, T)$ and $q : (W, R) \rightarrow (Y, S)$ are factor maps, in the topological sense.
- The map $f \mapsto \int f \, d\mu$ is a positive linear form on the algebra $\mathcal{B}$ and thus, there exists a unique probability measure ρ on W satisfying

$$\int f \, d\mu = \int \Psi(f) \, d\rho \quad \text{for all functions } f \in \mathcal{B}.$$

- As $\Psi(f \circ T) = \Psi(f) \circ R$ for all $f \in \mathcal{B}$ and μ is invariant under T , the measure ρ is invariant under R .
- As $\Psi(f) = f \circ p$ for all continuous functions f on X , we have that the image of ρ under p is equal to μ .
- Therefore, $p : (W, \rho, R) \rightarrow (X, \mu, T)$ is a measure theoretic factor map.
- * Moreover, for every function $f \in \mathcal{B}$,

$$\int |\Psi(f)|^2 \, d\rho = \int \Psi(|f|^2) \, d\rho = \int |f|^2 \, d\mu$$

and the map Ψ is an isometry from the space $\mathcal{B}$ endowed with the norm $L^2(\mu)$ into the space $L^2(\rho)$.

- Since $\mathcal{C}(X)$ is dense in $\mathcal{B}$ under the $L^2(\mu)$ norm and since $\Psi(f) = f \circ p$ for $f \in \mathcal{C}(X)$, we have that, for all $f \in \mathcal{B}$,

$$\Psi(f) = f \circ p \quad (p\text{-almost everywhere}).$$

- We claim that the map $p : (W, \rho, R) \rightarrow (X, \mu, T)$ is an isomorphism between measure preserving systems.
 - Indeed, the range of the map

$$f \mapsto f \circ p : L^2(\mu) \rightarrow L^2(\rho)$$

is closed in $L^2(\rho)$ because this map is an isometry, and it contains $\Psi(\mathcal{B}) = \mathcal{C}(W)$ and thus it is equal to $L^2(\rho)$.

- * In particular, (W, ρ, R) is ergodic.
- Finally, for every function $g \in \mathcal{C}(Y)$, we have that $g \circ q = \Psi(g \circ \pi) = g \circ \pi \circ p$ (ρ -almost everywhere) and so $q = \pi \circ p$ (ρ -almost everywhere).
 - In particular, the image of ρ under q is ν .

Subsection 6.1.2 (p. 40)

- Since ρ is ergodic under R , it admits a generic point w_1 .
 - Recall that this means that, for every $f \in \mathcal{C}(W)$,

$$\lim_{j \rightarrow +\infty} \frac{1}{j} \sum_{n=0}^{j-1} f(R^n w_1) = \int f \, d\rho.$$

- Set $x_1 = p(w_1)$.
- Since x_0 is a transitive point of X , we can choose,

□

2 Conclusion

Proposition 2.1 (cf. Host, 2019, Proposition 5). *Let (X, Γ) be a topological dynamical system where Γ is an amenable group, $x_0 \in X$, and μ be an ergodic invariant probability measure supported on the closed orbit of x_0 under the action of Γ .*

Let (Z, m_Z, S) be the Kronecker factor of (X, μ, T) , with factor map $p : X \rightarrow Z$.

Let $X \times Z$ be endowed with the group action of $\Gamma \times H$. Let $\tilde{\mu}$ be the measure on $X \times Z$ and image of μ under the map $X \rightarrow X \times Z$ where $x \mapsto (x, p(x))$.

Then there exists a Følner sequence $\tilde{\Phi}$ and a point $z_0 \in Z$ such that (x_0, z_0) is generic for $\tilde{\mu}$ along $\tilde{\Phi}$.

Proof. Let $d_X(\cdot, \cdot)$ and $d_Y(\cdot, \cdot)$ for distances on X and Y .

Let $\mathcal{B}$ be the closed subalgebra of $L^\infty(\mu)$.

□

3 Recent Publications

- Reflect on amenable result from newer paper within the wider context of the paper.
- Write own proof not using Gelfand spectrum.

Relatively recently, Charamaras and Mountakis (2024) proved analogous results for amenable groups:

Proposition 3.1 (Charamaras and Mountakis, 2024, Lemma 3.5). *Let Γ be an amenable group, let (X, μ, T) be an ergodic Γ -system, (Z, m_Z, S) be its Kronecker factor and $p : (X, \mu, T) \rightarrow (Z, m_Z, S)$ be a factor map.*

If $x \in X$ is a transitive point, then there exists a point $z \in Z$ and a Følner sequence Ψ' such that

$$\lim_{N \rightarrow \infty} \frac{1}{|\Psi'_N|} \sum_{\gamma \in \Psi'_N} f_1(T^\gamma x) \cdot f_2(S^\gamma z) = \int_X f_1 \cdot (f_2 \circ p) \, d\mu$$

holds for any $f_1 \in C(X)$ and $f_2 \in C(Z)$.

Proposition 3.2 (Charamaras and Mountakis, 2024, Proposition 3.7 (i & ii)). *Let Γ be an amenable group, let (X, μ, T) be an ergodic Γ -system, and let $x \in \text{gen}(\mu, \Phi)$ for some Følner sequence Φ . Let (Z, m_Z, S) be its Kronecker factor and $p : (X, \mu, T) \rightarrow (Z, m_Z, S)$ be a factor map, $X \times Z$ be endowed with the transformation $T \times S$, and $\tilde{\mu}$ be the joining of (X, μ, T) and (Z, m_Z, S) .*

Then there exists an ergodic extension $(X \times Z, \tilde{\mu}, T \times S)$ of (X, μ, Γ) , a Følner sequence $\tilde{\Phi}$, and a point $(x, z) \in \text{gen}(\tilde{\mu}, \tilde{\Phi})$ such that:

1. *The projection map π_X is a continuous factor map with $\pi_X(x, z) = x$.*
2. *The projection map π_Z is a continuous factor map where (Z, m_Z, S) is the Kronecker factor of $(X \times Z, \tilde{\mu}, T \times S)$.*

4 Summary

Theorem 4.1. *Let Γ be an amenable group, $A \subset \Gamma$, and Φ be a Følner sequence in Γ such that the limit*

$$\delta = \lim_{N \rightarrow \infty} \frac{|A \cap \Phi_N|}{|\Phi_N|}$$

exists.

Then there exists an ergodic system (X, μ, Γ) that is acted on by Γ , a factor map to its ergodic Kronecker factor $p : (X, \mu, T) \rightarrow (Z, m_Z, S)$, Følner sequences Ψ and $\tilde{\Phi}$ in Γ , and a point $(a, z_0) \in X \times Z$ such that:

1. *We have $z_0 = p(a)$.*
2. *We have clopen sets $E \subset X$ and $\tilde{E} \subset Z$ where $\tilde{E} = p(E)$ and*

$$\begin{aligned} A &= \{\gamma \in \Gamma : T^\gamma a \in E\} \\ &= \{\gamma \in \Gamma : (T \times S)^\gamma(a, z_0) \in \tilde{E}\}. \end{aligned}$$

3. *a is generic with μ with respect to Ψ and z_0 is generic with m_Z with respect to $\tilde{\Phi}$.*
4. *$\mu(E) = m_Z(\tilde{E}) \geq \delta$.*
5. *The factor map $p : (X, \mu, T) \rightarrow (Z, m_Z, S)$ is continuous.*

Remark 4.1. Kronecker factors are 1-step Pronilfactors and the continuous factor map $p : (X, \mu, T) \rightarrow (Z, m_Z, S)$ shows that a Kronecker factor is a topological pronilfactor.

5 JdLG-Decomposition

- Charamaras, D. and Mountakis, A. (2024). *Finding product sets in some classes of amenable groups*. <https://doi.org/https://doi.org/10.1017/fms.2024.155>.
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